Ch 1 · Cardiac arrest management

Core ALS

- High-quality CPR: 100–120/min, 5–6 cm, full recoil, pauses < 10 s, rotate every 2 min
- Shockable: shock → 2 min CPR; adrenaline 1 mg + amiodarone 300 mg after 3rd shock (150 mg after 5th)
- Non-shockable: adrenaline 1 mg ASAP, then every 3–5 min
- Airway: stepwise; waveform capnography for all intubated patients

Monitoring CPR quality

- **ETCO2** > 10–20 mmHg; sudden rise = ROSC; < 10 after 20 min = poor prognosis
- Arterial diastolic pressure > 25–35 mmHg (coronary perfusion surrogate)

Reversible causes and rescue

- 4 Hs (hypoxia, hypovolaemia, K+/metabolic, hypothermia) and 4 Ts (tension, tamponade, toxins, thrombosis)
- Focused echo in pauses; ECPR for selected refractory VF; PCI/thrombolysis when cause is clear

Key point: Minimise no-flow time; every answer starts with CPR quality and early defibrillation.

Ch 2 · Post-cardiac arrest care

Immediate priorities

- Coronary angiography: emergent if STEMI; consider if shock/electrical instability
- SpO2 94–98%, PaCO2 normal, MAP ≥ 65 (avoid hypotension), glucose 7.8–10 mmol/L
- Lung-protective ventilation; treat seizures (EEG); avoid hyperthermia

Temperature control

- Prevent fever (≤ 37.5 °C) for ≥ 72 h in comatose survivors
- TTM (2013) 33 vs 36 °C and TTM2 (2021) 33 °C vs normothermia: no difference
- HYPERION: possible benefit of 33 °C in non-shockable rhythms; treat shivering

Neuroprognostication (≥ 72 h, confounders excluded)

- Absent pupillary + corneal reflexes; bilaterally absent N20 SSEP
- Highly malignant EEG, status myoclonus ≤ 72 h, NSE > 60 µg/L, diffuse injury on CT/MRI
- Need ≥ 2 concordant predictors; avoid premature withdrawal

Key point: Post-arrest syndrome = brain injury + myocardial dysfunction + ischaemia–reperfusion + cause.

Ch 10 · Accidental hypothermia

Staging

- Mild 32–35 °C (shivering); moderate 28–32 °C (no shivering, AF, J waves); severe < 28 °C (coma, VF risk)
- Cold diuresis, coagulopathy, left-shifted O2 curve, slowed drug metabolism

Rewarming

- Passive external → active external (forced air) → active internal (warm fluids, lavage, ECLS)
- Arrest or unstable: **ECLS (VA-ECMO/bypass)** is the rewarming method of choice

Cardiac arrest specifics

- < 30 °C: up to 3 shocks, withhold adrenaline; 30–35 °C: double drug interval
- Mechanical CPR for transfer; K+ > 12 mmol/L suggests non-survivable; HOPE score for ECLS
- Complications: afterdrop, rewarming shock, arrhythmias, electrolyte shifts

Key point: "Not dead until warm and dead" – unless obvious lethal injury or very high K+.

Ch 11 · Cardiogenic shock

Definition and staging

- SBP < 90 (or vasopressor) + hypoperfusion + CI < 2.2 with raised filling pressure
- SCAI A–E (at risk → extremis); causes: AMI 80%, mechanical complications, myocarditis, cardiomyopathy

Management

- Early revascularisation (SHOCK); culprit-only PCI (CULPRIT-SHOCK)
- Noradrenaline first; dobutamine/milrinone for low output
- PA catheter: CPO = MAP × CO / 451 (< 0.6 W = poor prognosis); shock team

Mechanical support

- IABP: no mortality benefit (IABP-SHOCK II); keep for mechanical complications
- Impella CP improved survival in STEMI-CS (DanGer-Shock); routine VA-ECMO no benefit (ECLS-SHOCK)

Key point: Answer structure: define → stage → revascularise → drugs → monitor → MCS → escalate/ceiling.

Ch 12 · Acute heart failure

Clinical profiles

- Warm–wet (commonest), cold–wet, warm–dry, cold–dry
- Phenotypes: decompensated HF, hypertensive pulmonary oedema, RV failure, cardiogenic shock
- Find precipitant: ACS, arrhythmia, hypertension, infection, PE, non-adherence (CHAMPIT)

Acute treatment

- Oxygen to SpO2 > 90%; CPAP/NIV for pulmonary oedema (faster relief – 3CPO)
- IV nitrates for high BP; IV loop diuretic (DOSE: bolus = infusion); check natriuresis at 2–6 h
- Avoid routine morphine; inotropes only with hypoperfusion

Before discharge

- Quadruple GDMT: ARNI/ACEi, beta blocker, MRA, SGLT2 inhibitor
- Rapid up-titration (STRONG-HF); devices (ICD/CRT) when indicated

Key point: Decongest by natriuresis targets, not by weight alone.

Ch 13 · Acute coronary syndrome

STEMI

- Aspirin + P2Y12 inhibitor; primary PCI within 120 min, else lysis within 10 min
- Inferior MI: right-sided leads (V4R) – RV infarct: avoid nitrates, cautious fluids
- Bradycardia/AV block: atropine, pacing; O2 only if SpO2 < 90%

NSTE-ACS

- hs-troponin 0/1 h algorithm; GRACE score
- Immediate angiography (< 2 h) if shock, refractory pain, arrhythmia; early (< 24 h) if high risk

Mechanical complications

- VSR (new murmur, O2 step-up), papillary muscle rupture (acute MR), free wall rupture (tamponade)
- Others: LV thrombus/aneurysm, pericarditis, arrhythmias

Key point: New murmur + shock after MI = echo now for a mechanical complication.

Ch 14 · Cardiac tamponade

Physiology

- Pericardial pressure > chamber filling pressure; speed of accumulation matters most
- Pulsus paradoxus > 10 mmHg from ventricular interdependence

Diagnosis

- Beck's triad (hypotension, raised JVP, muffled sounds); low voltage, electrical alternans
- Echo: RA systolic and RV diastolic collapse, plethoric IVC, inflow variation (MV > 25%)

Management

- Urgent echo-guided pericardiocentesis; surgery for haemopericardium, dissection, loculated effusion
- Fluid bolus to temporise; avoid positive pressure ventilation where possible
- Send fluid for cytology/culture; window or drain for malignant recurrence

Key point: Intubating a tamponade patient can precipitate arrest – drain first if possible.

Ch 15 · Hypertensive crises

Definitions

- Emergency = severe BP + acute target-organ damage; urgency = no acute damage
- Organs: encephalopathy, stroke/ICH, ACS, pulmonary oedema, dissection, AKI, TMA, retinopathy

General treatment

- IV titratable agents (labetalol, nicardipine, clevidipine), arterial line
- MAP ↓ ≤ 25% in first hour, 160/100 by 2–6 h, normal over 24–48 h; avoid sublingual nifedipine

Special targets

- Aortic dissection: SBP < 120, HR < 60 within 20 min (beta blocker first)
- Ischaemic stroke: < 185/110 for lysis; otherwise treat only if > 220/120
- Pre-eclampsia: labetalol/hydralazine/nifedipine + magnesium; phaeochromocytoma: alpha before beta

Key point: Too fast a fall causes watershed ischaemia – except in dissection.

Ch 16 · AF and supraventricular tachycardias

New AF in ICU

- Triggers: sepsis, leak/surgery, low K/Mg, catecholamines, hypovolaemia, hypoxia
- Unstable due to AF → synchronised DC cardioversion
- Correct K 4–5, Mg > 1; rate control: amiodarone if hypotensive, beta blocker (esmolol/landiolol) if BP allows
- Anticoagulation by CHA2DS2-VASc vs bleeding risk; > 48 h needs anticoagulation/TOE before cardioversion

Other SVTs

- AVNRT/AVRT: vagal manoeuvres, adenosine 6–12–18 mg
- Flutter: sawtooth, 2:1 at 150/min – rate control, cardioversion, ablation
- MAT: treat lung disease, Mg; pre-excited AF: avoid AV-nodal blockers, use procainamide/DC

Key point: Treat the precipitant first – many ICU AF episodes revert spontaneously.

Ch 17 · Ventricular arrhythmias

Electrical storm

- ≥ 3 sustained VT/VF episodes (or ICD therapies) in 24 h
- Causes: scar re-entry, ischaemia, electrolytes, drugs, sympathetic drive, thyrotoxicosis

Management

- Unstable VT: cardioversion; VF: defibrillation; correct K+ and Mg2+
- Amiodarone ± lidocaine; propranolol for sympathetic blockade; deep sedation/intubation
- Interrogate ICD; exclude ischaemia; refractory: ablation, stellate ganglion block, MCS

Torsades de pointes

- Magnesium 2 g IV; stop QT-prolonging drugs; K+ > 4.5
- Increase rate: isoprenaline or overdrive pacing

Key point: Monomorphic VT with scar = re-entry; polymorphic VT = think ischaemia or long QT.

Ch 18 · Acute aortic syndromes

Classification

- Stanford A (ascending, surgical) vs B (distal to left subclavian); DeBakey I–III
- Includes dissection, intramural haematoma, penetrating ulcer

Management

- Esmolol/labetalol to HR < 60, then vasodilator to SBP 100–120; analgesia
- Type A: emergency surgery (mortality 1–2%/h untreated)
- Type B uncomplicated: medical; complicated (malperfusion, rupture, refractory pain/BP): TEVAR

Complications

- AR, tamponade, coronary occlusion, stroke, paraplegia
- Mesenteric, renal and limb malperfusion; rupture

Key point: Arm BP difference + tearing back pain + AR murmur = CT angiography now.

Ch 19 · Infective endocarditis

Duke criteria

- Major: typical organism in 2 cultures/persistent bacteraemia; vegetation, abscess, new regurgitation
- Minor: predisposition, fever, vascular and immunological phenomena, atypical micro
- Definite: 2 major, 1 major + 3 minor, or 5 minor

Treatment

- MSSA: flucloxacillin/cefazolin; MRSA: vancomycin/daptomycin; 4–6 weeks
- POET: partial oral therapy safe in stable left-sided IE

Surgery indications

- Heart failure from valve dysfunction; uncontrolled infection (abscess, persistent bacteraemia, fungi)
- Embolic prevention: vegetation > 10 mm with embolism or > 30 mm; Endocarditis Team

Key point: S. aureus bacteraemia always needs echo (TOE) and repeat cultures.

Ch 32 · Acute right ventricular failure

Pathophysiology

- Thin, afterload-sensitive RV; ↑ PVR → dilatation, TR, septal shift → low LV filling
- Hypotension → RV ischaemia → spiral of failure

Management

- Treat trigger; avoid hypoxia, hypercapnia, acidosis (raise PVR)
- Preload: usually diurese/ultrafiltrate; fluid only if clearly underfilled
- Keep systemic BP: noradrenaline, vasopressin; inotropes: dobutamine, milrinone
- Pulmonary vasodilators: inhaled NO/epoprostenol, IV prostacyclin; VA-ECMO as bridge

Intubation danger

- Induction vasodilation + positive pressure + apnoeic hypercapnia → arrest
- Pressors running, haemodynamically stable induction, low PEEP, ECMO standby

Key point: In RV failure, protect coronary perfusion: MAP above pulmonary pressure.

Ch 92 · Acute limb ischaemia

Rutherford classification

- I viable; IIa marginally threatened (minimal sensory loss); IIb immediately threatened (sensory + motor)
- III irreversible: paralysis, anaesthesia, rigor
- 6 Ps: pain, pallor, pulselessness, paraesthesia, paralysis, perishing cold

Management

- IV heparin immediately; analgesia
- IIb: urgent embolectomy or endovascular thrombectomy; IIa: catheter thrombolysis or surgery
- III: amputation or palliation

After revascularisation

- Compartment syndrome – fasciotomy; reperfusion hyperkalaemia, acidosis, myoglobinuria/AKI
- Find source (AF, LV thrombus) – echo and long-term anticoagulation

Key point: Sensory loss with weakness = hours matter.